



ARNAB BHOWMIK

RESEARCH SCHOLAR

PhD research scholar in theoretical physics at IEST Shibpur, working on ultracold quantum gases: Bose–Einstein condensates, quantum droplets and dipolar gases. Uses the extended Gross–Pitaevskii equation and Bogoliubov–de Gennes analysis to study how quantum fluctuations shape their stability and collective excitations. Also teaches physics and mathematics to school and undergraduate students.

EDUCATION

- Ph.D. in Physics** (ongoing) 2023 – Present
Indian Institute of Engineering Science and Technology (IEST), Shibpur, Howrah
Supervisor: Dr. Dwipesh Majumder, Associate Professor · Coursework CGPA 9.63 / 10
- M.Sc. in Physics** 2016 – 2018
Rajabazar Science College, University of Calcutta, Kolkata
First Class · CGPA 6.85 (60%) · Specialization: Quantum Electronics; Atomic and Molecular Physics
- B.Sc. in Physics** 2012 – 2015
Ramakrishna Mission Vivekananda Centenary College, Rahara · West Bengal State University
First Class · CGPA 7.25 (65%)

RESEARCH EXPERIENCE

- Senior Research Fellow (Physics)** Aug 2025 – Present
Junior Research Fellow (Physics) Aug 2023 – Aug 2025
Department of Physics, IEST Shibpur
- Theory of self-bound quantum droplets in spin-1 Bose–Einstein condensates in the polar phase, where Lee–Huang–Yang quantum fluctuations in the density and spin channels stop mean-field collapse.
 - Bogoliubov–de Gennes analysis of collective excitations: density and spin-wave modes in a quadratic Zeeman field; roton minimum, sound velocity and compressibility for finite-range (Pöschl–Teller) interactions.
 - Modelling of ground states and dynamics with the extended Gross–Pitaevskii equation.

EXPERTISE

THEORETICAL

- Beyond-mean-field (Lee–Huang–Yang) theory of quantum droplets
- Spinor and dipolar Bose–Einstein condensates
- Bogoliubov–de Gennes theory of collective excitations
- Finite-range (Pöschl–Teller) interactions; roton spectra and structure factor

COMPUTATIONAL / TECHNICAL

- Numerical modelling with the extended Gross–Pitaevskii equation
- Ground-state and real-time dynamics calculations
- [Add: programming languages]
- [Add: software and tools]

PUBLICATIONS

- [2] A. Bhowmik, D. Majumder, A. Banerjee. “Collective Excitations of Quantum Droplets in Spin-1 Bose-Einstein Condensate in Polar Phase.” *SSRN preprint* (2026). doi:10.2139/ssrn.7454627
- [1] A. Banerjee, A. Bhowmik, D. Majumder. “Collective excitation of Bose-Einstein condensate of Bose atoms with Pöschl-Teller interaction.” *Physics Letters A* **562**, 131006 (2025). doi:10.1016/j.physleta.2025.131006 · arXiv:2510.13383

CONFERENCES & WORKSHOPS

[Add: event, venue, month and year — talk, poster or participant]

AWARDS

Qualified GATE

2023

CONTACT

ADDRESS

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PROFILES

- Google Scholar
- LinkedIn
- ResearchGate

TEACHING

Physics & Mathematics Educator

2016 – Ongoing

Independent tutor for school and undergraduate students, with an emphasis on building intuition for difficult ideas.

REFEREES

Dwipesh Majumder

Associate Professor, Dept. of Physics,
IEST Shibpur (PhD supervisor)
dwipesh@gmail.com

Debnarayan Jana

Professor of Physics, University of
Calcutta
Google Scholar profile

Arka Chatterjee

Assistant Professor, UPES, Dehradun
Google Scholar profile

LANGUAGES

Bengali · English · Hindi

INTERESTS

- Geopolitics
- Expeditions and the exploration of remote places
- General knowledge
- Guitar (amateur)